CONJURA · NOTE

The Uber Assumption and its Random Self-Reducibility

in Non-Bilinear and Type 1, 2, 3 Bilinear Groups

Abstract

We fix notation for prime-order groups without a pairing and for bilinear groups in each of the three Galbraith–Paterson–Smart types, formally define the Uber assumption in all four settings, and treat random self-reducibility in each. A single generic-computability lemma isolates, per setting, the space of exponents a reduction can realize; a single affine-reparametrization theorem then yields random self-reducibility whenever the source data are stable under the reparametrization and the challenge polynomial transforms covariantly. The four settings differ only through the product structure the pairing (and, in Type 2, the homomorphism ψ) makes available, and this difference is exactly what enlarges or shrinks the self-reducible subclass. We give a worked self-reducible assumption in each setting, show that no monomial target can ever separate two settings, and exhibit a non-monomial target that is self-reducible in Type 2 but for which, in Type 3, the affine reduction fails at every worst-case point — the largest the failure locus can be, since the admissible reparametrizations form a group and the failure fraction is therefore either $O(1/p)$ or exactly one. Because the admissibility conditions are polynomial conditions on the reparametrization, the search for one is mechanical, and we give a decision procedure that returns the maximal admissible group and either certifies the reduction or exhibits the locus of worst-case points at which it fails. A further section places the theorem against classical random self-reducibility. The affine condition is the one-query, linear case of the non-adaptive condition of Feigenbaum and Fortnow; Lipton's multi-query line interpolation, which the group encoding admits unchanged, solves the computational problem for every target here, including the separating one, but only against a high-advantage solver and never the decisional problem. The classical decoding machinery that would soften its threshold is not linear in the received word and so does not survive the encoding, which is why the separation stands in the decisional, negligible-advantage regime that cryptography works in.

1 Introduction

The discrete-logarithm problem is random self-reducible by the shift $g^x \mapsto g^x g^r$ [TW87]; the answer for a worst-case instance is recovered from an average-case solver by subtracting r. This note carries the underlying idea — injecting a uniform reparametrization of the secret and undoing it on the answer — across the standard group settings, and states it uniformly. Throughout, p is a prime and all groups have order p. We use implicit (bracket) notation in the style of [EHKRV13]. Section 10 relates the resulting condition to the classical form of the same idea, in which the undoing is done by polynomial interpolation across several correlated queries rather than in closed form on one [Lipton91, FF93].

2 Group settings

2.1 Non-bilinear groups

Definition 1 (prime-order group). A *(non-bilinear) group generator* outputs $\mathcal{G} = (p, \mathbb{G}, g)$ where $\mathbb{G}$ is a cyclic group of prime order p, g a generator, and the group law and equality are computable in probabilistic polynomial time. For $a \in \mathbb{Z}_p$ write $[a] \coloneqq g^a$, so $[a] \cdot [b] = [a+b]$ and $[a]^c = [ca]$. There is no pairing.

2.2 Bilinear groups and the GPS taxonomy

Definition 2 (bilinear group). A *bilinear group generator* outputs $\mathcal{BG} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, g_1, g_2)$ where $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ are cyclic of prime order p, g_1, g_2 generate $\mathbb{G}_1, \mathbb{G}_2$, all group laws and equalities are efficient, and $e\colon \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ is an efficiently computable, non-degenerate bilinear map. For $a \in \mathbb{Z}_p$ write $[a]_1 \coloneqq g_1^a$, $[a]_2 \coloneqq g_2^a$, and $[a]_T \coloneqq e(g_1, g_2)^a$, so that

$$e\big([a]_1, [b]_2\big) = [ab]_T, \qquad [a]_i \cdot [b]_i = [a+b]_i, \qquad [a]_i^c = [ca]_i.$$

Following Galbraith, Paterson, and Smart [GPS08], bilinear groups come in three types, distinguished by the efficiently computable isomorphisms between $\mathbb{G}_1$ and $\mathbb{G}_2$:

Type 1. $\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}$ and e is symmetric, $e(u,v) = e(v,u)$. One source group.

Type 2. $\mathbb{G}_1 \neq \mathbb{G}_2$, with an efficiently computable isomorphism $\psi\colon \mathbb{G}_2 \to \mathbb{G}_1$ (so $\psi([a]_2) = [a]_1$), but none known in the reverse direction.

Type 3. $\mathbb{G}_1 \neq \mathbb{G}_2$, with no efficiently computable isomorphism in either direction.

Type 3 is the most efficient and most used in practice; Type 1 is the most permissive algebraically. The only structural difference for what follows is which group elements can be moved between $\mathbb{G}_1$ and $\mathbb{G}_2$, and hence which products the pairing can form.

3 The generic-computability calculus

Fix a setting and an instance consisting of encodings of finite polynomial tuples $R \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_1$, $S \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_2$, and $T \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_T$ (in a single-group setting only R and, if applicable, T occur). Each tuple is assumed to contain the constant 1, since the generators are public. For $\vec{X} = (X_1, \dots, X_n)$ and a secret $\vec{x} = (x_1, \dots, x_n) \in \mathbb{Z}_p^n$, the instance is the collection of $[r_i(\vec{x})]_1$, $[s_j(\vec{x})]_2$, $[t_k(\vec{x})]_T$.

For a group $\gamma \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$ (or $\mathbb{G}$) let $\mathcal{L}_\gamma \subseteq \mathbb{Z}_p[\vec{X}]$ denote the polynomials q for which $[q(\vec{x})]_\gamma$ is computable in probabilistic polynomial time from the instance and public parameters, uniformly in $\vec{x}$. Write $A \cdot B = \{ab : a \in A,\, b \in B\}$ and let $\langle\cdot\rangle$ be $\mathbb{Z}_p$-linear span.

Lemma 1 (computable exponents). *With the conventions above,*

$$\begin{aligned}
\textit{non-bilinear:}\quad & \mathcal{L}_{\mathbb{G}} = \langle R\rangle;\\
\textit{Type 1 } (\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G},\ \textit{source tuple } R):\quad & \mathcal{L}_{\mathbb{G}} = \langle R\rangle,\quad \mathcal{L}_{\mathbb{G}_T} = \langle R\cdot R\ \cup\ T\rangle;\\
\textit{Type 2 } (\psi\colon \mathbb{G}_2 \to \mathbb{G}_1):\quad & \mathcal{L}_1 = \langle R\cup S\rangle,\quad \mathcal{L}_2 = \langle S\rangle,\\
& \mathcal{L}_T = \langle (R\cup S)\cdot S\ \cup\ T\rangle;\\
\textit{Type 3:}\quad & \mathcal{L}_1 = \langle R\rangle,\quad \mathcal{L}_2 = \langle S\rangle,\quad \mathcal{L}_T = \langle R\cdot S\ \cup\ T\rangle.
\end{aligned}$$

The inclusions "$\supseteq$" hold unconditionally: every listed polynomial's encoding is computable by group operations and pairings. The reverse inclusions hold in the generic bilinear group model, where these are exactly the computable encodings [BBG05, Boyen08].

Justification. The group law realizes all $\mathbb{Z}_p$-linear combinations, giving $\langle\cdot\rangle$ in each group. The pairing sends a computable $\mathbb{G}_1$-exponent $a \in \mathcal{L}_1$ and a computable $\mathbb{G}_2$-exponent $b \in \mathcal{L}_2$ to $ab \in \mathcal{L}_T$; pairing against the generator ($b = 1$ or $a = 1$) shows $\mathcal{L}_1, \mathcal{L}_2 \subseteq \mathcal{L}_T$. In Type 1 both pairing arguments range over $\langle R\rangle$, producing all products $R\cdot R$. In Type 2, ψ moves each $\mathbb{G}_2$ encoding into $\mathbb{G}_1$, so $\mathcal{L}_1$ gains S, and the pairable products become $\mathcal{L}_1\cdot\mathcal{L}_2 = (R\cup S)\cdot S$; the missing products $R\cdot R$ would require an element of $\mathbb{G}_2$ with an R-exponent, which ψ cannot supply. In Type 3 neither group maps to the other, so $\mathcal{L}_1 = \langle R\rangle$, $\mathcal{L}_2 = \langle S\rangle$, and only cross products $R\cdot S$ arise. The generic-model converse is the master lemma of [BBG05, Boyen08]. □

The single distinction across the four rows is the set of $\mathbb{G}_T$-products: none (no pairing), $R\cdot R$ (Type 1), $R\cdot S\cup S\cdot S$ (Type 2), $R\cdot S$ (Type 3). Everything below is a consequence of this.

4 The Uber assumption

Definition 3 (bilinear Uber problem). Fix a bilinear setting, source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$), each containing 1, a challenge polynomial $f \in \mathbb{Z}_p[\vec{X}]$, and a challenge group $\mathsf{ch} \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$; write $\mathcal{L}_\star := \mathcal{L}_{\mathsf{ch}}$ (Lemma 1). For a secret $\vec{x} \in \mathbb{Z}_p^n$ the instance is

$$\mathsf{Inst}(\vec{x}) = \big(\{[r_i(\vec{x})]_1\}_i,\ \{[s_j(\vec{x})]_2\}_j,\ \{[t_k(\vec{x})]_T\}_k\big).$$

- *Computational* $(R, S, T, f; \mathsf{ch})$-Uber: given $\mathsf{Inst}(\vec{x})$, output $[f(\vec{x})]_\star$.
- *Decisional* $(R, S, T, f; \mathsf{ch})$-Uber: given $\mathsf{Inst}(\vec{x})$ and Z, decide whether $Z = [f(\vec{x})]_\star$ (real) or $Z \leftarrow_\$ \mathsf{ch}$ (random).

The *average case* draws $\vec{x} \leftarrow_\$ \mathbb{Z}_p^n$; the *worst case* quantifies over all $\vec{x}$. For a decision algorithm $\mathcal{A}$ put

$$\delta(\mathcal{A}; \vec{x}) = \Pr\big[\mathcal{A}(\mathsf{Inst}(\vec{x}), [f(\vec{x})]_\star) = 1\big] - \Pr\big[\mathcal{A}(\mathsf{Inst}(\vec{x}), U) = 1\big],\quad U \leftarrow_\$ \mathsf{ch},$$

and $\delta_{\mathrm{avg}}(\mathcal{A}) = \mathbb{E}_{\vec{x}\leftarrow_\$\mathbb{Z}_p^n}[\delta(\mathcal{A}; \vec{x})]$; for a search algorithm put $\mathsf{Succ}(\mathcal{A}; \vec{x}) = \Pr[\mathcal{A}(\mathsf{Inst}(\vec{x})) = [f(\vec{x})]_\star]$ and $\mathsf{Succ}_{\mathrm{avg}}(\mathcal{A}) = \mathbb{E}_{\vec{x}}[\mathsf{Succ}(\mathcal{A}; \vec{x})]$. The assumption asserts that δ_{avg} (resp. $\mathsf{Succ}_{\text{avg}}$) is negligible for all probabilistic polynomial-time $\mathcal{A}$.

The four settings are Definition 3 with the computable spaces of Lemma 1. We record each explicitly.

Definition 4 (non-bilinear Uber). Setting $\mathcal{G} = (p, \mathbb{G}, g)$; source tuple R (in $\mathbb{G}$, containing 1); challenge $f \in \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}$. Instance $\mathsf{Inst}(\vec{x}) = \{[r_i(\vec{x})]\}_i$; goal, compute or distinguish $[f(\vec{x})]$. Computable space $\mathcal{L}_\star = \mathcal{L}_{\mathbb{G}} = \langle R \rangle$; no pairing, hence no products.

Definition 5 (Type-1 Uber). Setting $\mathcal{BG}$ of Type 1, $\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}$; source tuple R (in $\mathbb{G}$), target tuple T (in $\mathbb{G}_T$), each containing 1; challenge f in $\mathrm{ch} \in \{\mathbb{G}, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_{\mathbb{G}} = \langle R \rangle$ and $\mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \cup T \rangle$.

Definition 6 (Type-2 Uber). Setting $\mathcal{BG}$ of Type 2 with $\psi\colon \mathbb{G}_2 \to \mathbb{G}_1$; source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$); challenge f in $\mathrm{ch} \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_1 = \langle R \cup S \rangle$, $\mathcal{L}_2 = \langle S \rangle$, $\mathcal{L}_T = \langle (R \cup S) \cdot S \cup T \rangle$.

Definition 7 (Type-3 Uber). Setting $\mathcal{BG}$ of Type 3; source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$); challenge f in $\mathrm{ch} \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_1 = \langle R \rangle$, $\mathcal{L}_2 = \langle S \rangle$, $\mathcal{L}_T = \langle R \cdot S \cup T \rangle$.

Remark 1 (hardness). The master theorem of [BBG05, Boyen08] states, informally, that the decisional $(R, S, T, f; \mathrm{ch})$-Uber problem is hard in the generic bilinear group model if and only if $f \notin \mathcal{L}_\star$, that is, the challenge exponent is linearly independent of the computable ones. Random self-reducibility is a separate property and does not require hardness; the two interact through Lemma 1, as discussed in Section 12.

5 An affine random-self-reducibility theorem

Let $\mathrm{AGL}_n(\mathbb{Z}_p)$ be the affine group of maps $\tau_{A,\vec{b}}\colon \vec{x} \mapsto A\vec{x} + \vec{b}$ with $A \in \mathrm{GL}_n(\mathbb{Z}_p)$, acting on polynomials by $(q \circ \tau)(\vec{X}) = q(A\vec{X} + \vec{b})$. The following theorem is setting-agnostic: it refers only to the computable spaces of Lemma 1.

Theorem 1 (master reduction). *Consider any of the four settings, an instance with source tuples R, S, T and challenge $(f; \mathrm{ch})$, and computable spaces $\{\mathcal{L}_\gamma\}$ with $\mathcal{L}_\star = \mathcal{L}_{\mathrm{ch}}$. Suppose there is an efficiently sampleable distribution $\mathcal{D}$ on a subgroup $\mathcal{T} \leq \mathrm{AGL}_n(\mathbb{Z}_p)$ such that, for every $\tau \in \mathcal{T}$,*

(C1) *$r_i \circ \tau \in \mathcal{L}_1$, $s_j \circ \tau \in \mathcal{L}_2$, and $t_k \circ \tau \in \mathcal{L}_T$ for all i, j, k (with $\mathcal{L}_1 = \mathcal{L}_{\mathbb{G}}$ in the single-group settings);*

(C2) *$f \circ \tau = \lambda_\tau f + h_\tau$ for a scalar $\lambda_\tau \in \mathbb{Z}_p^\times$ and a polynomial $h_\tau \in \mathcal{L}_\star$, both computable from τ and the parameters;*

and, moreover,

(C3) *$\tau(\vec{x})$ is uniform on $\mathbb{Z}_p^n$ for every fixed $\vec{x}$ when $\tau \leftarrow \mathcal{D}$.*

Then the computational and decisional $(R, S, T, f; \mathrm{ch})$-Uber problems are random self-reducible: there is a probabilistic polynomial-time reduction $\mathcal{B}^{\mathcal{A}}$, using one draw from $\mathcal{D}$ and one oracle call, with $\mathsf{Succ}(\mathcal{B}; \vec{x}_0) = \mathsf{Succ}_{\mathrm{avg}}(\mathcal{A})$ and $\delta(\mathcal{B}; \vec{x}_0) = \delta_{\mathrm{avg}}(\mathcal{A})$ for every $\vec{x}_0 \in \mathbb{Z}_p^n$.

Proof. On input $\mathsf{Inst}(\vec{x}_0)$ (and a challenge Z in the decisional case) the reduction proceeds as follows.

Reparametrize. Draw $\tau \leftarrow \mathcal{D}$ and set $\vec{x}' := \tau(\vec{x}_0)$, which is unknown to $\mathcal{B}$ but uniform on $\mathbb{Z}_p^n$ by (C3).

Rebuild the instance. By (C1), each rebuilt exponent lies in the computable space of its group, so by the "$\supseteq$" part of Lemma 1 its encoding is computable from $\mathsf{Inst}(\vec{x}_0)$. Concretely, writing $r_i \circ \tau = \sum_a c_a g_a$ as a $\mathbb{Z}_p$-combination of the generators of $\mathcal{L}_1$, one has $[r_i(\vec{x}')]_1 = [(r_i \circ \tau)(\vec{x}_0)]_1 = \prod_a [g_a(\vec{x}_0)]_1^{c_a}$, and similarly for S (using ψ in Type 2 to realize any S-exponent in $\mathbb{G}_1$) and for T (using pairings $e([\cdot]_1, [\cdot]_2)$ to realize products). This yields $\mathsf{Inst}(\vec{x}')$, distributed exactly as a fresh average-case instance.

Transport the challenge. By (C2) compute λ_τ, h_τ and the element $[h_\tau(\vec{x}_0)]_\star$ (computable since $h_\tau \in \mathcal{L}_\star$).

Decisional case. Set $Z' := Z^{\lambda_\tau} \cdot [h_\tau(\vec{x}_0)]_\star$ and output $\mathcal{A}(\mathsf{Inst}(\vec{x}'), Z')$. If $Z = [f(\vec{x}_0)]_\star$ then

$$Z' = [\lambda_\tau f(\vec{x}_0) + h_\tau(\vec{x}_0)]_\star = [(f \circ \tau)(\vec{x}_0)]_\star = [f(\vec{x}')]_\star,$$

the real challenge for $\vec{x}'$. If Z is uniform on ch, then Z^{λ_τ} is uniform because $z \mapsto z^{\lambda_\tau}$ is a bijection of the order-p cyclic group ch (as $\gcd(\lambda_\tau, p) = 1$), so Z' is uniform and independent of $\mathsf{Inst}(\vec{x}')$. Writing p_1, p_0 for $\mathcal{A}$'s acceptance probabilities on real and random challenges,

$$\delta(\mathcal{B}; \vec{x}_0) = \mathbb{E}_\tau[p_1(\tau(\vec{x}_0))] - \mathbb{E}_\tau[p_0(\tau(\vec{x}_0))] \overset{(C3)}{=} \mathbb{E}_{\vec{x}'}[p_1] - \mathbb{E}_{\vec{x}'}[p_0] = \delta_{\mathrm{avg}}(\mathcal{A}).$$

Search case. Run $W \leftarrow \mathcal{A}(\mathsf{Inst}(\vec{x}'))$. On success $W = [f(\vec{x}')]_\star = [\lambda_\tau f(\vec{x}_0) + h_\tau(\vec{x}_0)]_\star$, so output $Y := (W \cdot [h_\tau(\vec{x}_0)]_\star^{-1})^{\lambda_\tau^{-1} \bmod p} = [f(\vec{x}_0)]_\star$. Since $\vec{x}'$ is uniform, the success probability equals $\mathsf{Succ}_{\mathrm{avg}}(\mathcal{A})$. □

Remark 2 (relaxing (C3)). When $\mathcal{D}$ makes $\tau(\vec{x})$ uniform only on a subset $\Omega \subseteq \mathbb{Z}_p^n$ (for instance $\mathbb{Z}_p^\times \times \mathbb{Z}_p$ under a multiplicative reparametrization of one coordinate), the transfer holds with the average taken over Ω. If the complement $\mathbb{Z}_p^n \setminus \Omega$ is a proper affine subvariety on which $f(\vec{x})$ takes a value the reduction can compute directly from $\mathsf{Inst}(\vec{x}_0)$ (a common case is $f(\vec{x}) = 0$ there), the reduction detects that case by an equality test on the given encodings and answers it without an oracle call, recovering full worst-case coverage.

6 Random self-reducibility: non-bilinear groups

Here $\mathcal{L}_\star = \mathcal{L}_{\mathbb{G}} = \langle R \rangle$ and no pairing exists, so (C2) demands the covariance defect to be a $\mathbb{Z}_p$-combination of the source exponents themselves. The motivating scalar case is discrete log: to solve g^x query the solver on g^{x+r} and subtract r [TW87]; its group-valued cousins fit Definition 3.

Proposition 1 (CDH and DDH). *Let $R = (1, X_1, X_2)$, so $\mathcal{L}_{\mathbb{G}} = \langle 1, X_1, X_2\rangle$, and $f = X_1X_2$ with challenge in $\mathbb{G}$. The translation group $\mathcal{T} = \{\vec{x} \mapsto \vec{x} + \vec{b}\}$ with $\vec{b} \leftarrow_\$ \mathbb{Z}_p^2$ satisfies* (C1)–(C3)*, so computational CDH and decisional DDH are random self-reducible.*

Proof. $X_i \circ \tau = X_i + b_i \in \mathcal{L}_{\mathbb{G}}$ gives (C1). For (C2), $f \circ \tau = (X_1 + b_1)(X_2 + b_2) = f + (b_2X_1 + b_1X_2 + b_1b_2)$, so $\lambda_\tau = 1$ and $h_\tau = b_2X_1 + b_1X_2 + b_1b_2 \in \langle 1, X_1, X_2\rangle = \mathcal{L}_\star$. Translation gives exact uniformity, (C3). Explicitly the reduction returns $[f(\vec{x}_0)] = W \cdot [x_1]^{-b_2}[x_2]^{-b_1}[1]^{-b_1b_2}$ in the search case. □

With $R = (1, X_1, \ldots, X_n)$ we have $\mathcal{L}_\star = V_{\le 1}$ (affine polynomials), and translating a degree-d monomial leaves monomials of degree at most $d-1$; hence (C2) holds under translation exactly when $\deg f \le 2$. Without a pairing, degree 2 is the ceiling: a product target is the most that is randomizable, and it is.

7 Random self-reducibility: Type 1

Here $\mathcal{L}_\star = \mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \cup T\rangle$: the pairing supplies all products of source exponents, so the covariance defect may use any degree-2 combination of them.

Proposition 2 (BDDH). *Let $R = (1, X_1, X_2, X_3)$, $T = (1)$, so $\mathcal{L}_{\mathbb{G}_T} = V_{\le 2}$, and $f = X_1X_2X_3$ with challenge in $\mathbb{G}_T$. Translation satisfies* (C1)–(C3)*, so decisional and computational BDDH are random self-reducible.*

Proof. $X_i \circ \tau = X_i + b_i \in \mathcal{L}_{\mathbb{G}} = \langle R\rangle$ gives (C1). Expanding,

$$f \circ \tau = X_1X_2X_3 + \left[b_3X_1X_2 + b_2X_1X_3 + b_1X_2X_3 + (\deg \le 1)\right],$$

so $\lambda_\tau = 1$ and h_τ has total degree at most 2, hence lies in $V_{\le 2} = \mathcal{L}_\star$; each of its monomials is realized by a single pairing, for instance $[x_1x_2]_T = e([x_1]_1, [x_2]_2)$. Translation gives (C3). □

With coordinate source $R = (1, X_1, \ldots, X_n)$ we get $\mathcal{L}_\star = V_{\le 2}$, so the same degree count gives translation-randomizability for every f with $\deg f \le 3$. The pairing has raised the ceiling from 2 (no pairing) to 3.

8 Random self-reducibility: Type 2

Here $\mathcal{L}_1 = \langle R \cup S\rangle$ and $\mathcal{L}_T = \langle (R \cup S) \cdot S \cup T\rangle$: relative to Type 3 the homomorphism ψ adds the right-hand products $S \cdot S$ to the target space. This extra room is what makes the following target randomizable.

Proposition 3. *In a Type-2 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, $T = (1)$, and $f = X_1X_2^2$ with challenge in $\mathbb{G}_T$. Then $f \notin \mathcal{L}_\star$ (so the assumption is generically hard), and translation satisfies* (C1)–(C3)*; the problem is random self-reducible.*

Proof. Here $\mathcal{L}_T = \langle (R \cup S) \cdot S \rangle = \langle 1, X_1, X_2, X_1X_2, X_2^2 \rangle$, and $X_1X_2^2$ is not in this span, giving generic hardness. Under $\tau\colon \vec{x} \mapsto \vec{x} + \vec{b}$ one has $X_1 \circ \tau = X_1 + b_1 \in \mathcal{L}_1 = \langle 1, X_1 \rangle$ and $X_2 \circ \tau = X_2 + b_2 \in \mathcal{L}_2 = \langle 1, X_2 \rangle$, which is (C1). Expanding,

$$f \circ \tau = (X_1 + b_1)(X_2 + b_2)^2 = f + \underbrace{2b_2X_1X_2 + b_2^2X_1 + b_1X_2^2 + 2b_1b_2X_2 + b_1b_2^2}_{h_\tau},$$

so $\lambda_\tau = 1$ and every monomial of h_τ lies in $\mathcal{L}_\star$. The term X_2^2 is realized using ψ: $[x_2^2]_T = e\big(\psi([x_2]_2), [x_2]_2\big)$. Translation gives (C3). □

The point is precisely the X_2^2 term of h_τ: it is a product of two $\mathbb{G}_2$ exponents, which the pairing can form only after ψ has moved one of them into $\mathbb{G}_1$. In Type 1 this term is free; in Type 3 it is unavailable, as the next section shows.

9 Random self-reducibility: Type 3

Here $\mathcal{L}_1 = \langle R \rangle$, $\mathcal{L}_2 = \langle S \rangle$, and $\mathcal{L}_T = \langle R \cdot S \cup T \rangle$: the target space contains only cross products. This is the most restrictive setting, and reparametrizations must respect which coordinates are reachable in the challenge group.

Proposition 4 (co-CDH). *In a Type-3 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, and $f = X_1X_2$ with challenge in $\mathbb{G}_1$. Then $f \notin \mathcal{L}_\star$ (generically hard), and the reparametrization $\tau\colon (x_1, x_2) \mapsto (\alpha x_1,\ x_2 + b_2)$ with $\alpha \leftarrow_\$ \mathbb{Z}_p^\times$, $b_2 \leftarrow_\$ \mathbb{Z}_p$ satisfies* (C1),(C2) *and makes $\tau(\vec{x})$ uniform on $\mathbb{Z}_p^\times \times \mathbb{Z}_p$; the problem is random self-reducible, with the slice $x_1 = 0$ handled directly.*

Proof. Here $\mathcal{L}_\star = \mathcal{L}_1 = \langle 1, X_1 \rangle$, which contains no $\mathbb{G}_2$ variable; and $X_1X_2 \notin \langle 1, X_1 \rangle$, giving hardness. For (C1), $X_1 \circ \tau = \alpha X_1 \in \mathcal{L}_1$ and $X_2 \circ \tau = X_2 + b_2 \in \mathcal{L}_2 = \langle 1, X_2 \rangle$. For (C2), $f \circ \tau = \alpha X_1(X_2 + b_2) = \alpha f + \alpha b_2 X_1$, so $\lambda_\tau = \alpha \in \mathbb{Z}_p^\times$ and $h_\tau = \alpha b_2 X_1 \in \mathcal{L}_\star$. The image $\tau(\vec{x}) = (\alpha x_1, x_2 + b_2)$ is uniform on $\mathbb{Z}_p^\times \times \mathbb{Z}_p$ for every fixed $x_1 \neq 0$. If $x_1 = 0$ then $f(\vec{x}) = 0$ and the answer is the identity of $\mathbb{G}_1$; the reduction detects this by testing $[x_1]_1 = [0]_1$ and outputs the identity, per Remark 2. A multiplicative reparametrization of x_1 is forced because an additive shift $x_1 \mapsto x_1 + b_1$ would put b_1X_2 into h_τ, and $X_2 \notin \mathcal{L}_1$ in Type 3. □

9.1 Monomial targets never separate

It is tempting to look for a separation at the target $f = X_1X_2^2$ of Proposition 3, whose defect under translation contains the X_2^2 that Type 3 cannot form. That attempt fails, and it fails for a reason that rules out an entire class of candidates at once.

Proposition 5 (monomial targets are always self-reducible). *Let R, S, T consist of monomials, each containing 1, and let $f = X_1^iX_2^j$ with $i, j \geq 1$ be a monomial with $f \notin \mathcal{L}_\star$. Then in every one of the four settings the diagonal torus $\mathcal{T} = \{\tau_{a,d}\colon \vec{x} \mapsto (ax_1, dx_2) : a, d \in \mathbb{Z}_p^\times\}$, with a and d drawn uniformly from $\mathbb{Z}_p^\times$, satisfies* (C1) *and* (C2)*, and Theorem 1 applies in the relaxed form of Remark 2. No such f separates two settings.*

Proof. A monomial composed with a coordinate scaling is a scalar multiple of itself, so $m \circ \tau_{a,d} = a^k d^l m$ for $m = X_1^k X_2^l$. Hence every source monomial stays in the span of itself, giving (C1) in any setting, and $f \circ \tau_{a,d} = a^i d^j f$, giving (C2) with $\lambda_\tau = a^i d^j \in \mathbb{Z}_p^\times$ and $h_\tau = 0$ — the defect vanishes identically, so no uncomputable monomial can appear in it. The image $\tau_{a,d}(\vec{x})$ is uniform on $\mathbb{Z}_p^\times \times \mathbb{Z}_p^\times$ whenever $x_1, x_2 \neq 0$, and on the complement $\{x_1 = 0\} \cup \{x_2 = 0\}$ one has $f(\vec{x}) = 0$ because $i, j \geq 1$, so the answer is the identity and the reduction detects the case by testing $[x_1]_1$ and $[x_2]_2$ against the identity. Remark 2 then gives full worst-case coverage. □

In particular $f = X_1 X_2^2$ is random self-reducible in Type 3 as well as in Type 2, by $\tau \colon \vec{x} \mapsto (\alpha x_1,\ x_2 + b)$: here $f \circ \tau = \alpha f + 2\alpha b\, X_1 X_2 + \alpha b^2 X_1$, whose defect monomials $X_1 X_2$ and X_1 are both cross products $R \cdot S$ and hence in $\mathcal{L}_T$ even in Type 3. This is the same shape of reparametrization that Proposition 4 forces for co-CDH. A separating target must therefore be non-monomial, and must defeat the torus.

9.2 A separating target

Proposition 6 (a Type-2 versus Type-3 separation). *Let $p \geq 5$, $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, $T = (1)$, and*

$$f = X_1^2 + X_1 X_2^2 + X_2^3$$

with challenge in $\mathbb{G}_T$. Then $f \notin \mathcal{L}_T$ in Type 2 and in Type 3, so the assumption is generically hard in both. In Type 2 the admissible reparametrizations act transitively and uniformly on $\mathbb{Z}_p^2$, so Theorem 1 applies with exact advantage transfer. In Type 3 the admissible group is

$$\mathcal{T}_3 = \{\tau_c \colon \vec{x} \mapsto (x_1 - 3c,\ x_2 + c)\ :\ c \in \mathbb{Z}_p\} \cong (\mathbb{Z}_p, +),$$

of order p; its orbits are the p lines $x_1 + 3x_2 = k$, each of size p; and on no such line is f computable from the instance. Hence (C3) *fails at* every *$\vec{x}_0 \in \mathbb{Z}_p^2$, even in the relaxed form of Remark 2.*

Proof. Hardness. In Type 3, $\mathcal{L}_T = \langle R \cdot S \rangle = \langle 1, X_1, X_2, X_1 X_2 \rangle$; in Type 2, $\mathcal{L}_T = \langle (R \cup S) \cdot S \rangle = \langle 1, X_1, X_2, X_1 X_2, X_2^2 \rangle$. Neither contains X_1^2, so $f \notin \mathcal{L}_T$ in either.

Type 3. By (C1), $\tau_1 = aX_1 + c_1$ and $\tau_2 = dX_2 + c_2$ with $a, d \in \mathbb{Z}_p^\times$. The monomials of $f \circ \tau$ lying outside $\mathcal{L}_T$ are X_1^2, $X_1 X_2^2$, X_2^3 and X_2^2; the first three occur in f and the fourth does not. Reading off coefficients,

$$[X_1^2] :\ a^2 = \lambda_\tau, \qquad [X_1 X_2^2] :\ ad^2 = \lambda_\tau, \qquad [X_2^3] :\ d^3 = \lambda_\tau.$$

From the first two, $a = d^2$; from the last two, $a = d$; hence $d^2 = d$ and, as $d \in \mathbb{Z}_p^\times$, $d = 1$, so $a = 1$ and $\lambda_\tau = 1$. The pure X_2^2 coefficient is $c_1 + 3c_2$ and must vanish, giving $c_1 = -3c_2$. Writing $c \coloneqq c_2$ yields exactly $\mathcal{T}_3$, and one checks $h_{\tau_c} = 2c\, X_1 X_2 + (c^2 - 6c) X_1 - 3c^2 X_2 + (9c^2 - 2c^3) \in \mathcal{L}_T$, so (C2) does hold on $\mathcal{T}_3$. Since $x_1 + 3x_2$ is invariant under τ_c, the orbit of $\vec{x}_0$ is the line $\ell_k = \{x_1 + 3x_2 = k\}$ through it, of size p out of p^2.

No escape on an orbit. Parametrize ℓ_k by $s \coloneqq x_2$, so $x_1 = k - 3s$. There

$$f|_{\ell_k} = (k - 3s)^2 + (k - 3s)s^2 + s^3 = -2s^3 + (k + 9)s^2 - 6ks + k^2,$$

while $\mathcal{L}_T$ restricts to $\langle 1,\, k-3s,\, s,\, (k-3s)s\rangle = \langle 1, s, s^2\rangle$. As $p \ge 5$ the coefficient -2 is nonzero and s^3 is not a $\mathbb{Z}_p$-combination of $1, s, s^2$ as a function on $\mathbb{Z}_p$, so $f|_{\ell_k} \notin \mathcal{L}_T|_{\ell_k}$ for every k. The escape clause of Remark 2 is therefore unavailable on any orbit.

Type 2. Now $\mathcal{L}_1 = \langle 1, X_1, X_2\rangle$, so (C1) permits $\tau_1 = a_{11}X_1 + a_{12}X_2 + c_1$, and $X_2^2 \in \mathcal{L}_T$ removes the X_2^2 constraint. The surviving conditions are $a_{11}^2 = \lambda_\tau$, $a_{11}d^2 = \lambda_\tau$ and $a_{12}d^2 + d^3 = \lambda_\tau$, giving $a_{11} = d^2$ and $a_{12} = d^2 - d$, with c_1, c_2 free. Then $\tau(\vec{x}) = (d^2x_1 + (d^2-d)x_2 + c_1,\ dx_2 + c_2)$, and as c_1, c_2 range over $\mathbb{Z}_p$ the two coordinates range independently over all of $\mathbb{Z}_p$: $\tau(\vec{x})$ is exactly uniform on $\mathbb{Z}_p^2$ for every fixed $\vec{x}$, which is (C3) unrelaxed. □

Two features of Type 2 drive the separation, and the proof uses both: ψ places X_2^2 in $\mathcal{L}_T$, which removes the constraint that ties c_1 to c_2 in Type 3, and it places X_2 in $\mathcal{L}_1$, which lets τ_1 carry the shear term $a_{12}X_2$. Neither is available in Type 3, and either alone would leave a smaller admissible group.

On the other side, all three terms of f are needed, and the coefficient conditions $a^2 = ad^2 = d^3 = \lambda_\tau$ are what make them work together: any two of the three force only a single relation between a and d, leaving a one-parameter torus, whereas all three force $d = 1$ and kill the torus that Proposition 5 would otherwise use to rescue the Type-3 reduction. Dropping a term accordingly costs the separation, but in different ways. Without X_1^2 one has $a = d$; the torus survives, and the failure locus shrinks to the single line $x_1 + 3x_2 = 0$, an $O(1/p)$ fraction. Without $X_1X_2^2$ one has $a^2 = d^3$, and without X_2^3 one has $a = d^2$; in both cases the surviving one-parameter torus leaves Type-3 orbits of size $\Theta(p^2)$, so the loss is a constant factor rather than a break — Type 2 retains full coverage in each case, so what fails is the Type-3 half of the separation, not the Type-2 half.

The hypothesis $p \ge 5$ is not cosmetic. The proof needs s^3 to be independent of $1, s, s^2$ *as a function* on $\mathbb{Z}_p$, which holds exactly when $p > 3$. At $p = 3$ Fermat gives $s^3 = s$, so $f|_{\ell_k} = ks^2 + s + k^2$ lies in $\mathcal{L}_T|_{\ell_k}$ and the escape clause of Remark 2 becomes available on every orbit; at $p = 2$ the assumption is vacuous, since X_1^2, X_2^2 and X_2^3 collapse to X_1, X_2, X_2 and f agrees pointwise with $X_1 + X_1X_2 + X_2 \in \mathcal{L}_T$. Both failures are of hardness or of the escape clause, not of the group computation, which is characteristic-free.

Remark 3 (the failure is as large as it can be). The set of τ satisfying (C1) and (C2) is closed under composition and inversion, so it is a group and the orbits of $\mathbb{Z}_p^n$ under it partition the space. Orbit size is constant on orbits, so the set of $\vec{x}_0$ at which (C3) fails is a union of orbits. If it is not all of $\mathbb{Z}_p^n$, it omits some orbit of size at least $p^n - O(p)$ and therefore has size $O(p)$. The failure fraction is thus either $O(1/p)$ or exactly 1, with nothing in between, and Proposition 6 realizes the second alternative.

Proposition 6 rules out the single-reparametrization technique of Theorem 1, which is the tool behind every positive result above. It does not rule out every reduction: Section 10 shows that a multi-query interpolation reduction still solves the *computational* problem, at high solver advantage, for this target as for any other with these source tuples. The Type-2 versus Type-3 gap established here is accordingly a gap for decisional reductions and for reductions that must work at negligible advantage. Ruling out every decisional reduction would additionally require a lower-bound argument such as a meta-reduction.

10 Relation to classical random self-reducibility

Random self-reducibility long predates its use on group-based assumptions, and the classical technique differs from Theorem 1 in one respect that decides exactly what Proposition 6 does and does not rule out. This section isolates the difference and delimits the reach of each technique.

10.1 The multi-query condition

Condition (C2) asks a single reparametrized challenge to be a scalar multiple of the original up to a computable defect. Feigenbaum and Fortnow [FF93] call a problem non-adaptively random self-reducible when $f(\vec{x}_0)$ is recoverable by an arbitrary polynomial-time function of the answers on several identically distributed queries; see [BT06] for the place of that notion in average-case complexity. The specialization in which the recovery is $\mathbb{Z}_p$-linear is the one this setting supports:

(C2′) there are $\tau_1, \ldots, \tau_m$, jointly sampled so that each τ_i is marginally distributed as in (C3), together with coefficients $c_1, \ldots, c_m \in \mathbb{Z}_p$ and a polynomial $h \in \mathcal{L}_\star$, all computable from the sampled data, such that

$$f = \sum_{i=1}^{m} c_i\,(f \circ \tau_i) + h.$$

Condition (C2) is the case $m = 1$ with $c_1 \neq 0$: from $f \circ \tau = \lambda_\tau f + h_\tau$ one gets $f = \lambda_\tau^{-1}(f \circ \tau) - \lambda_\tau^{-1} h_\tau$, which is (C2′) with $c_1 = \lambda_\tau^{-1}$ and $h = -\lambda_\tau^{-1} h_\tau$.

Given (C1) for each τ_i, condition (C2′) suffices for the *computational* problem: the reduction rebuilds $\mathsf{Inst}(\tau_i(\vec{x}_0))$ for each i as in the proof of Theorem 1, queries the solver on all m of them to get $W_1, \ldots, W_m$, and returns

$$\prod_{i=1}^{m} W_i^{c_i} \cdot [h(\vec{x}_0)]_\star = \Big[\sum_{i=1}^{m} c_i\, f(\tau_i(\vec{x}_0)) + h(\vec{x}_0)\Big]_\star = [f(\vec{x}_0)]_\star.$$

The step that makes this go through is that a $\mathbb{Z}_p$-linear combination of *exponents* is realized by group operations on their encodings, so linear post-processing of the solver's answers is available to the reduction for free. Encodings are transparent to it: the map $\vec{x} \mapsto [f(\vec{x})]_\star$ is a Reed–Muller codeword read through the encoding, and a linear local-decoding procedure for that code never notices the encoding is there. This is why the linear part of the classical toolkit — Lipton's line restriction [Lipton91], and the degree-k curves with which Beaver and Feigenbaum [BF90] make the queries k-wise independent — carries over to computational Uber problems unchanged.

10.2 Interpolation, and what it costs

The classical instantiation restricts f to a random line through $\vec{x}_0$ and interpolates. It needs nothing of f beyond its degree, and nothing of the setting beyond (C1) for translations.

Proposition 7 (line interpolation). *Suppose R, S and T are stable under translation, in the sense that $r \circ \tau_{\vec{b}} \in \mathcal{L}_1$, $s \circ \tau_{\vec{b}} \in \mathcal{L}_2$ and $t \circ \tau_{\vec{b}} \in \mathcal{L}_T$ for every $\vec{b} \in \mathbb{Z}_p^n$ and every $r \in R$, $s \in S$, $t \in T$, where $\tau_{\vec{b}} \colon \vec{x} \mapsto \vec{x} + \vec{b}$. Let $d := \deg f$ and suppose $d \le p - 2$. Then the* computational $(R, S, T, f; \mathrm{ch})$*-Uber problem is random self-reducible by a non-adaptive $(d+1)$-query reduction $\mathcal{B}^{\mathcal{A}}$ with*

$$\mathsf{Succ}(\mathcal{B}; \vec{x}_0) \;\ge\; 1 - (d+1)\big(1 - \mathsf{Succ}_{\mathrm{avg}}(\mathcal{A})\big) \qquad \textit{for every } \vec{x}_0 \in \mathbb{Z}_p^n.$$

Proof. Draw $\vec{b} \leftarrow_{\$} \mathbb{Z}_p^n$ and fix $d+1$ distinct nonzero $t_1, \ldots, t_{d+1} \in \mathbb{Z}_p$, which exist as $d + 1 \le p - 1$. Put $\tau_i := \tau_{t_i \vec{b}}$. Each τ_i is a translation, so (C1) holds by hypothesis and $\mathsf{Inst}(\vec{x}_0 + t_i \vec{b})$ is computable from $\mathsf{Inst}(\vec{x}_0)$; and since $t_i \ne 0$ and $\vec{b}$ is uniform, $\vec{x}_0 + t_i \vec{b}$ is uniform on $\mathbb{Z}_p^n$, so each query is marginally a fresh average-case instance. The $d+1$ queries are not jointly independent: they lie on one line.

Put $g(t) := f(\vec{x}_0 + t\vec{b})$, a polynomial in t of degree at most d, and let $c_i := \prod_{j \ne i} (-t_j)(t_i - t_j)^{-1}$ be the Lagrange coefficients at 0, so that $q(0) = \sum_i c_i q(t_i)$ for every q of degree at most d. This is (C2′) with $m = d+1$ and $h = 0$. Query $W_i \leftarrow \mathcal{A}(\mathsf{Inst}(\vec{x}_0 + t_i \vec{b}))$ and output $Y := \prod_i W_i^{c_i}$; if every $W_i = [g(t_i)]_\star$ then $Y = [\sum_i c_i g(t_i)]_\star = [g(0)]_\star = [f(\vec{x}_0)]_\star$. A union bound over the $d+1$ queries gives the stated bound. □

The source tuples of Propositions 4, 5 and 6 are all translation-stable, so Proposition 7 applies to each: the computational problems there are self-reducible in Type 3 whatever the covariance obstruction, including the separating target of Proposition 6, whose degree 3 needs four queries. The reduction never rebuilds the covariance defect — the solver returns it implicitly inside its own answers — so no uncomputable monomial ever has to be formed.

What it costs is the regime. The queries must be correlated for the interpolation identity to hold, so the transfer runs through a union bound and the bound is vacuous unless $\mathsf{Succ}_{\mathrm{avg}}(\mathcal{A}) > d/(d+1)$. In the classical setting that threshold is soft: Berlekamp–Welch decoding brings the tolerable disagreement to any constant below $1/2$ [GS92], and list decoding brings it past $1/2$ [Sudan97, STV01]. Neither is available here. Berlekamp–Welch solves $N(t_i) = y_i E(t_i)$ for the coefficients of N and E, a linear system whose *coefficient matrix contains the received values y_i*; the reduction holds only $[y_i]_\star$, and forming that system would require multiplying and inverting encoded values. The same objection applies to the list-decoding refinements. So the very criterion that lets interpolation survive the encoding — linearity in the received word — excludes the machinery that would soften its threshold, and $d/(d+1)$ is a genuine wall rather than an artifact.

Remark 4 (challenge transport). Interpolation does not reach the decisional problem. Recovering $f(\vec{x}_0)$ from the answers is linear and hence free in the exponent, but a decisional reduction must also run in the other direction, transporting its own challenge Z forward to a challenge for each rebuilt instance, and decision bits do not interpolate. Concretely, for the target $f = X_1^2 + X_1X_2^2 + X_2^3$ of Proposition 6 the reduction would need the coefficients of $g(t) = f(\vec{x}_0 + t\vec{b})$; the coefficient of t is the directional derivative

$$b_1(2X_1 + X_2^2) + b_2(2X_1X_2 + 3X_2^2) = 2b_1X_1 + 2b_2X_1X_2 + (b_1 + 3b_2)X_2^2,$$

whose X_2^2 term lies outside $\mathcal{L}_T$ in Type 3 unless $b_1 + 3b_2 = 0$ — that is, unless $\vec{b}$ points along the one direction whose orbits are the degenerate lines of Proposition 6.

10.3 What each technique buys

The two techniques are not ordered by reach. On *computational* problems interpolation is the more permissive of the two: it asks nothing of f but its degree, so wherever the sources are translation-stable it succeeds regardless of the covariance obstruction, and Proposition 7 applies to every target in this note. What separates them is the regime. Theorem 1 transfers advantage exactly, and reaches the decisional problem; Proposition 7 says nothing below solver advantage $d/(d+1)$ and nothing about decisional problems at all (Remark 4). The gap is not one of tightness — a reduction requiring $\mathsf{Succ}_{\mathrm{avg}} > 3/4$ is not a loose reduction, it does not exist below that threshold — and, by the preceding subsection, it cannot be closed by the classical decoding machinery, because that machinery is not linear in the received word. Since Uber assumptions are stated at negligible advantage, and very often in decisional form, the covariance route is in practice the only one available; that is why Theorem 1, and not interpolation, is the tool used throughout this note, and why the separation of Proposition 6 is meaningful despite Proposition 7.

Two further points of contact are worth recording, with their differences. Shift-based self-correction in the style of Blum, Luby and Rubinfeld [BLR93] uses the same $\vec{x} \mapsto \vec{x} + \vec{b}$ as the CDH and DDH reduction of Section 6, but their corrector queries the oracle twice per trial and takes a majority over repetitions, tolerating a constant error rate, where the reduction here makes one query and computes the correction itself from the instance, transferring advantage exactly. And the worst-case-to-average-case reductions of lattice cryptography, from Ajtai [Ajtai96] to Regev [Regev09], share the motivation of this note — basing an average-case assumption on a worst-case one — but are reductions *between distinct problems* rather than self-reductions, and rest on lattice geometry rather than on polynomial structure; Regev's is moreover a quantum reduction.

11 Deciding whether the affine technique applies

Conditions (C1) and (C2) are polynomial conditions on the entries of τ, and (C3) is a statement about the size of an orbit. Finding an admissible $\mathcal{T}$ is therefore a computation rather than a matter of ingenuity, and the hand arguments of Sections 6–9 are all instances of one procedure. This section writes it out. What the procedure decides is the applicability of Theorem 1, not random self-reducibility as such; Section 11.3 is precise about the gap.

11.1 Set-up

Fix a setting and an instance $(R,S,T,f;\mathrm{ch})$ in n variables over $\mathbb{Z}_p$, and put $D :=\max\{\deg f,\ \deg q : q \in R\cup S\cup T\}$. Affine substitution does not raise degree, so every polynomial the procedure meets lies in $P_D := \mathbb{Z}_p[\vec{X}]_{\le D}$, a $\mathbb{Z}_p$-space of dimension $N = \binom{n+D}{n}$; represent its elements by coefficient vectors. Treat the n^2+n entries of $(A,\vec{b})$ in $\tau_{A,\vec{b}}\colon \vec{x}\mapsto A\vec{x}+\vec{b}$ as indeterminates; for $q\in P_D$ the coefficients of $q\circ\tau_{A,\vec{b}}$ are then polynomials in those indeterminates, obtained by expansion.

Two structural facts fix the shape of the procedure. First, (C1) and (C2) cut out an algebraic subset $\mathcal{T}\subseteq \mathrm{AGL}_n(\mathbb{Z}_p)$ which is a subgroup (Remark 3). Second, with $\mathcal{D}$ uniform on $\mathcal{T}$ the image $\tau(\vec{x}_0)$ is uniform on the orbit $\mathcal{T}\vec{x}_0$, because $\tau\mapsto\tau(\vec{x}_0)$ is $|\mathrm{Stab}_{\mathcal{T}}(\vec{x}_0)|$-to-one onto that orbit. Deciding (C3) is therefore deciding how large the orbits are, and by the Lang–Weil estimate an orbit of dimension e carries $\Theta(p^e)$ points, with implied constants depending only on n and D. For p large relative to those, comparing $\dim\mathcal{T}\vec{x}_0$ with n decides (C3) up to exactly the $O(1/p)$ slack that Remark 2 already tolerates.

11.2 The procedure

Admissible$(\Gamma,R,S,T,f,\mathrm{ch},p)$ — decides applicability of Theorem 1.
Returns one of Not-Hard, Exact, Relaxed, Affine-Fails, together with $\mathcal{T}$ and, in the last case, a witness locus.

1. *Computable spaces.* From Lemma 1 form the generating sets for the setting Γ and compute bases of $\mathcal{L}_1,\mathcal{L}_2,\mathcal{L}_T\subseteq P_D$ by Gaussian elimination. Put $\mathcal{L}_\star := \mathcal{L}_{\mathrm{ch}}$.
2. *Hardness.* Test $f\in\mathcal{L}_\star$ by linear algebra. If so, return Not-Hard: the challenge is already computable from the instance and there is no assumption to reduce.
3. *Admissible group.* Let $\pi\colon P_D\to P_D/\mathcal{L}_\star$ and $\bar{f} := \pi(f)$, nonzero by step 2. Impose on $(A,\vec{b})$:
   (a) (C1) for each $r\in R$, $s\in S$, $t\in T$, the vanishing of the image of $\mathrm{coef}(r\circ\tau)$ in $P_D/\mathcal{L}_1$, of $\mathrm{coef}(s\circ\tau)$ in $P_D/\mathcal{L}_2$, and of $\mathrm{coef}(t\circ\tau)$ in $P_D/\mathcal{L}_T$;
   (b) (C2) the vanishing of every 2×2 minor of the two-column matrix $[\,\pi(f\circ\tau)\mid\bar{f}\,]$.

   Let I be the ideal they generate. Compute a Gröbner basis of I saturated at $\det A$ and at $\pi(f\circ\tau)$; the resulting variety is $\mathcal{T}$, and the scalar λ_τ is read off from $\pi(f\circ\tau)=\lambda_\tau\bar{f}$. If $\mathcal{T}$ is trivial, return Affine-Fails with witness $\mathbb{Z}_p^n$.
4. *Generic orbit.* Compute $e := \dim\mathcal{T}\vec{x}_0$ for generic $\vec{x}_0$, by eliminating $(A,\vec{b})$

from $= A\vec{x}_0 + \vec{b}$ over I. If $e < n$, return AFFINE-FAILS with witness $\mathbb{Z}_p^n$: by Remark 3 every orbit is then deficient, so (C3) fails everywhere.

5. *Exactness.* If $\mathcal{T}$ acts transitively — equivalently $|\mathcal{T}\vec{x}_0| = p^n$ for one, hence by the dichotomy for every, $\vec{x}_0$ — return EXACT: Theorem 1 applies with unrelaxed (C3) and exact advantage transfer.

6. *Failure locus.* Otherwise let $Z := \{\vec{x}_0 : \dim \mathcal{T}\vec{x}_0 < n\}$, the locus where the evaluation map drops rank; obtain its irreducible components $Z_1, \ldots, Z_m$ by primary decomposition. By Remark 3, Z is a proper closed subset, so $|Z| = O(p^{n-1})$.

7. *Escape clause.* For each component Z_j test both:

   (a) *detectability* — is Z_j cut out by equations $q(\vec{x}) = 0$ with $q \in \mathcal{L}_\gamma$ for some group γ? If so the reduction decides membership by testing $[q(\vec{x}_0)]_\gamma$ against the identity;

   (b) *computability* — writing ρ_j for restriction to Z_j, is $\rho_j(f) \in \rho_j(\mathcal{L}_\star)$? This is linear algebra in the coordinate ring of Z_j, and must be done on *functions* on $Z_j(\mathbb{Z}_p)$, not on polynomials.

   If both hold for every j, return RELAXED: Theorem 1 applies via Remark 2, with advantage transfer up to $O(1/p)$. Otherwise return AFFINE-FAILS with the first offending Z_j as witness.

Proposition 8 (correctness). *ADMISSIBLE terminates, and its output is correct: it returns NOT-HARD exactly when $f \in \mathcal{L}_\star$; it returns EXACT only when some efficiently sampleable $\mathcal{D}$ satisfies* (C1)–(C3) *as stated in Theorem 1; it returns RELAXED only when Theorem 1 applies in the form of Remark 2; and when it returns AFFINE-FAILS with witness Z_j, no distribution on any subgroup of* $\mathrm{AGL}_n(\mathbb{Z}_p)$ *satisfies* (C1)–(C3) *at the points of Z_j, even relaxed.*

Proof. Termination is that every step is a finite linear-algebra or Gröbner computation over $\mathbb{Z}_p$. Step 2 is the generic-model hardness criterion of [BBG05, Boyen08]. For steps 3–5, the conditions imposed are literal transcriptions of (C1) and (C2): the former because membership of $q \circ \tau$ in a subspace is the vanishing of its image in the quotient, the latter because $f \circ \tau = \lambda_\tau f + h_\tau$ with $h_\tau \in \mathcal{L}_\star$ says exactly that $\pi(f \circ \tau)$ is proportional to $\bar{f}$, the factor being λ_τ and the saturation enforcing $\lambda_\tau \neq 0$. Since $\mathcal{T}$ is a group and $\mathcal{D}$ is uniform on it, $\tau(\vec{x}_0)$ is uniform on $\mathcal{T}\vec{x}_0$; so (C3) holds exactly when the orbit is all of $\mathbb{Z}_p^n$ and holds to within $O(1/p)$ when the orbit has full dimension. Maximality is what gives the negative half: $\mathcal{T}$ contains *every* τ meeting (C1) and (C2), so any admissible subgroup is contained in $\mathcal{T}$ and its orbits are contained in $\mathcal{T}$-orbits; if the $\mathcal{T}$-orbit of a point of Z_j is already deficient and f is not computable on it, no smaller group can do better. ◻

Remark 5 (cost). The expensive step is 3, whose Gröbner computation is doubly exponential in $n^2 + n$ in the worst case. The systems arising here are far from worst case: they are the coefficient-matching equations of Section 9, near-triangular in the entries of A, and for $n = 2$ they are six variables and are solved by hand. Steps 1, 2 and 7 are $O(N^3)$ linear algebra with $N = \binom{n+D}{n}$.

11.3 What the procedure does not decide

AFFINE-FAILS is not a proof that the assumption fails to be random self-reducible. It is a proof about one technique, and three gaps remain open beneath it.

First, multi-query reductions. Proposition 7 solves the computational problem whenever the sources are translation-stable and $\deg f \leq p - 2$, whatever the verdict of ADMISSIBLE — as it does for the separating target of Proposition 6. A run returning AFFINE-FAILS should therefore be followed by that test, which is a one-line check. The general linear multi-query condition (C2′) is decidable too, but by a different route: it asks whether f lies in $\langle f \circ \tau : \tau \text{ satisfies (C1)} \rangle + \mathcal{L}_\star$, a subspace of P_D that can be accumulated by sampling τ until the dimension stabilises, which happens after at most N successful samples. That test is necessary but not sufficient, since (C2′) also requires the sampled τ_i to be marginally distributed as in (C3).

Second, non-linear recombination. Feigenbaum and Fortnow [FF93] allow the answers to be combined by an arbitrary polynomial-time function, and nothing here decides that condition; the procedure sees only the $\mathbb{Z}_p$-linear fragment that the group encoding makes free.

Third, reparametrizations outside $\mathrm{AGL}_n(\mathbb{Z}_p)$, and reductions that are not reparametrization based at all. A genuine impossibility result for the decisional problem would need a lower-bound argument such as a meta-reduction, as noted after Proposition 6.

ADMISSIBLE is accordingly a decision procedure for a sufficient condition, and a semi-decision procedure for random self-reducibility: it can certify it, and it cannot refute it.

12 Discussion

Across the four settings the reduction of Theorem 1 is one and the same; only the admissibility conditions move, and they move only because the computable target space $\mathcal{L}_\star$ grows with the available products:

$$\text{none} \subsetneq R \cdot S \text{ (Type 3)} \subsetneq R \cdot S \cup S \cdot S \text{ (Type 2)} \subseteq R \cdot R \text{ (Type 1, single group)}.$$

A larger $\mathcal{L}_\star$ makes the covariance condition (C2) easier to meet, so the affine-self-reducible subclass is largest in Type 1 and smallest in Type 3, with the non-bilinear case the degenerate product-free floor. Proposition 6 witnesses the strict gap between Type 2 and Type 3, at a target where the Type-3 reduction fails at every worst-case point — the most the failure locus can ever be, by Remark 3. The gap is one for decisional reductions and for reductions that must work at negligible advantage: Proposition 7 still solves the computational problem in Type 3 at high solver advantage, as it does for every target here. Proposition 5 marks the other boundary, that no monomial target can separate any two settings at all.

Self-reducibility is orthogonal to hardness. The master theorem of [BBG05, Boyen08] makes an assumption hard exactly when $f \notin \mathcal{L}_\star$, whereas Theorem 1 needs the defect $f \circ \tau - \lambda_\tau f$ to lie *in* $\mathcal{L}_\star$. These constrain different objects, so hard yet self-reducible assumptions abound: CDH and DDH (non-bilinear), BDDH (Type 1), and co-CDH and every monomial target (Type 3, Proposition 5) are all of this kind. Quantitatively, Bauer, Fuchsbauer, and Loss [BFL20] show that in the algebraic group model every member of the Uber family is implied by the q-discrete-logarithm assumption, with q governed by the degrees of the defining polynomials, and Rotem [Rotem22] refines q to the maximal degree in which a single variable occurs and gives improved bilinear reductions.

13 Bibliography

[TW87] M. Tompa and H. Woll. Random self-reducibility and zero knowledge interactive proofs of possession of information. In *28th Annual Symposium on Foundations of Computer Science (FOCS 1987)*, pages 472–482. IEEE, 1987.

[BF90] D. Beaver and J. Feigenbaum. Hiding instances in multioracle queries. In *7th Annual Symposium on Theoretical Aspects of Computer Science (STACS 1990)*, volume 415 of *Lecture Notes in Computer Science*, pages 37–48. Springer, 1990.

[Lipton91] R. J. Lipton. New directions in testing. In *Distributed Computing and Cryptography*, volume 2 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 191–202. AMS, 1991.

[GS92] P. Gemmell and M. Sudan. Highly resilient correctors for polynomials. *Information Processing Letters*, 43(4):169–174, 1992.

[BLR93] M. Blum, M. Luby, and R. Rubinfeld. Self-testing/correcting with applications to numerical problems. *Journal of Computer and System Sciences*, 47(3):549–595, 1993. Preliminary version in *STOC 1990*, pages 73–83.

[FF93] J. Feigenbaum and L. Fortnow. Random-self-reducibility of complete sets. *SIAM Journal on Computing*, 22(5):994–1005, 1993.

[Ajtai96] M. Ajtai. Generating hard instances of lattice problems. In *28th Annual ACM Symposium on Theory of Computing (STOC 1996)*, pages 99–108. ACM, 1996.

[Sudan97] M. Sudan. Decoding of Reed–Solomon codes beyond the error-correction bound. *Journal of Complexity*, 13(1):180–193, 1997.

[STV01] M. Sudan, L. Trevisan, and S. Vadhan. Pseudorandom generators without the XOR lemma. *Journal of Computer and System Sciences*, 62(2):236–266, 2001.

[BBG05] D. Boneh, X. Boyen, and E.-J. Goh. Hierarchical identity based encryption with constant size ciphertext. In *Advances in Cryptology – EUROCRYPT 2005*, volume 3494 of *Lecture Notes in Computer Science*, pages 440–456. Springer, 2005.

[BT06] A. Bogdanov and L. Trevisan. Average-case complexity. *Foundations and Trends in Theoretical Computer Science*, 2(1):1–106, 2006.

[Boyen08] X. Boyen. The uber-assumption family: a unified complexity framework for bilinear groups. In *Pairing-Based Cryptography – Pairing 2008*, volume 5209 of *Lecture Notes in Computer Science*, pages 39–56. Springer, 2008.

[GPS08] S. D. Galbraith, K. G. Paterson, and N. P. Smart. Pairings for cryptographers. *Discrete Applied Mathematics*, 156(16):3113–3121, 2008.

[Regev09] O. Regev. On lattices, learning with errors, random linear codes, and cryptography. *Journal of the ACM*, 56(6):article 34, 2009. Preliminary version in *STOC 2005*, pages 84–93.

[EHKRV13] A. Escala, G. Herold, E. Kiltz, C. Ràfols, and J. Villar. An algebraic framework for Diffie-Hellman assumptions. In *Advances in Cryptology – CRYPTO 2013*, volume 8043 of *Lecture Notes in Computer Science*, pages 129–147. Springer, 2013. Journal version: *Journal of Cryptology* 30(1):242–288, 2017.

[BFL20] B. Bauer, G. Fuchsbauer, and J. Loss. A classification of computational assumptions in the algebraic group model. In *Advances in Cryptology – CRYPTO 2020*, volume 12171 of *Lecture Notes in Computer Science*, pages 121–151. Springer, 2020.

[Rotem22] L. Rotem. Revisiting the uber assumption in the algebraic group model: fine-grained bounds in hidden-order groups and improved reductions in bilinear groups. In *3rd Conference on Information-Theoretic Cryptography (ITC 2022)*, volume 230 of *LIPIcs*, article 13. Schloss Dagstuhl, 2022.